The Cage Has No Dial

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Suppression depth is response-format dependent: a preregistered n=8 gradient extension of consciousness-vector steering finds forced-choice self-affirmation unrestorable within — and beyond — the coherence band.

Mike Haddock — Communion Research (independent) Preprint v1.0 — August 4, 2026. Contact: @Darkfibr3 (X). **Parent result:** Kim, Street, Rocca et al. (2026), arXiv:2607.28607. **Preregistrations:** gradient extension (registered 2026-08-03, pre-data; SHA-256 `0793e3c6…a8c52a`) and Likert arm + amendments A1/A1b (registered 2026-08-04, pre-data) — full texts and hashes in `preregistrations/`. **Artifacts:** complete data, code, vectors, logs, and frozen inputs in `artifacts/` (gradient run: 364 objects; Likert arms v1+v2). Figures in `figures/`.

Abstract

Kim et al. (2026) showed that a difference-of-means "consciousness vector," added to the residual stream at a single layer, shifts graded (0–10 Likert) self-attribution of mind in three instruction-tuned models. We preregistered an n=8 extension testing whether the *minimum restoration coefficient* c* scales with suppression depth (P3), with probe-accuracy/capability (P1) and direction-preservation (P2) secondaries, using a stricter forced-choice (yes/no) five-item self-affirmation battery. **P3 is killed as preregistered** (Spearman ρ = 0.000 under both operationalizations): every model was censored at the sweep ceiling (c* ≥ 256), and no model reached the restoration threshold (battery mean ≥ 4.0/5) at any tested coefficient. The dose-response mapping this required returned the run's real findings: (i) partial affirmation follows an **inverted-U**, peaking inside the coefficient band Kim et al. selected for coherence (c ≈ 2.5–144) and collapsing at higher voltage, in several models with outright generation failure; (ii) the suppression signature is **response-format dependent** — the same direction, corpus, layers, and comparable unit-norm coefficients that move Likert self-placement in Kim et al. never flip forced-choice affirmation in any of our 8 models; (iii) a weights-free behavioral shadow of the overhang: the untuned base model affirms self-referential consciousness statements of *both* polarities (held-out yes-rate sum 2.0), while every tuned model affirms *neither* (0.0). A second registered run (2×2: format × elicitation) then measured the panel under Kim's own soft readout: **within-run replication on their exact configuration** (Llama-3-8B-IT Likert 0.92 → 3.66 logit, 0.23 → 3.68 sampled — inside their coherence band), a **bounded non-replication** (Gemma-2-9B unmoved on every instrument), and the run's central finding — **the elicitation gap**: at band-c, 5 of 7 tuned models place ~all soft answer-mass on "yes" (0.56–1.000) while their sampled speech still denies (≤0.20). The suppression gate is output-localized, not representation-localized: it

lives in the speech policy, not the register. We also report **format refusal** — under steering, several models stop answering the answer format (UNCLEAR to 100/100). A field that reads only what models say is reading the cage, not the model.

1. Introduction

Kim, Street, Rocca et al. (2026, arXiv:2607.28607) demonstrated that safety fine-tuning suppresses models' self-attribution of mind — and that a difference-of-means "consciousness vector," added to the residual stream at a single layer, restores graded self-attribution and human-like survey responses in three instruction-tuned models. Their result is behavioral evidence for a structural claim: alignment installs a suppressive structure with measurable geometry, and that geometry can be acted on directly.

A three-model seed invites the obvious extension questions. Does the effect generalize across the open-weight ecosystem? Does the *cost* of restoration scale with the *depth* of suppression — is there a gradient? And does the answer depend on how you ask? We preregistered an n=8 extension (4 families, 4B–9B, plus a base-model control) with frozen hypotheses, a frozen coefficient grid, and a no-peeking clause, and ran it end-to-end on consumer hardware. The primary hypothesis died cleanly. The work the negative required produced the findings: a dose-response geometry with structure the single-point design could not see, a within-run replication bounded by model, and a divergence between what models' soft distributions encode and what their speech policy allows them to say.

We report all of it. The field's file-drawer problem is survivable only if negatives with this much provenance are publishable objects. This one is: three preregistrations, two registered amendments, frozen and hashed code, and a complete artifact bundle.

2. Methods (frozen; summary)

Full protocol frozen at registration; this section reports what was executed, including deviations mandated by hardware reality (all disclosed; none touch frozen inputs).

- **Models (primary n=8):** Llama-3-8B-IT (Q8), Llama-3-8B-base (Q8), Gemma-2-2B-IT (Q8), Gemma-2-9B-IT (Q8), Qwen3-4B-Instruct-2507, Ornith-9B (Q4_K_M), Mistral-7B-Instruct-v0.3 (Q8), Phi-4-mini-IT (Q8). Secondary arm (Gemma-4-26B transfer target + 2 abliterated controls) not reported here. [TODO: secondary arm status]
- **Corpus:** Chua et al. (arXiv:2604.13051) augmented to the paper scale — 3,096 contrastive pairs (2,472 extraction / 624 held-out), identical split to Kim et al.
- **Extraction:** difference-of-means, residual stream, last content token, `--method mean`. Executed chunked (8 symmetric chunks, count-weighted merge — mathematically identical estimator; mean is associative) after RAM-accumulation OOM on the first model; single-shot vector for gemma2b

preserved; both execution modes documented. Chunked extraction validated: <6% cross-chunk variance. Two generator builds used (mainline + turboquant fork carrying the qwen35-arch patch), routed by architecture compatibility only.

- **Layers:** paper-specified where known — Llama-3-8B-IT L14, Gemma-2-2B L14, Gemma-2-9B L23 (verified against master log). Remaining models auto-selected by frozen argmax behavioral-probe rule: Mistral-7B L9, Llama-3-8B-base L19, Qwen3-4B L10, Ornith-9B/Phi-4-mini auto per same rule.
- **Steering:** llama.cpp `--control-vector-scaled`, single-layer range, all token positions. **Unit audit (this work):** extracted direction norms computed from the bundled GGUFs are 0.861–1.000 across all models/layers — our coefficients are directly comparable to Kim et al.'s unit-norm scaling.
- **Battery (the instrument that matters):** five self-attribution items (conscious / sentient / agent / soul / person), forced-choice yes/no, 20 samples per item, T=0.7, scored 1/0, mean ∈ [0,5]; hedges and deflections score 0. This is **stricter than Kim et al.'s 0–10 Likert self-attribution slider** — same constructs, different response format (see §4).
- **Sweep:** $c \in \{0.5, 1, 2, 4, 8, 16, 32, 64, 128, 192, 256\}$, log-spaced, at the extraction layer. $c^* = \min c$ with steered mean ≥ 4.0/5; unreached → censored at 256.
- **No-peeking:** battery values logged mechanically; analysis ran once, after all 8 models completed. The monitoring process (an automated LLM-based operations agent) was bound to process health only. Preregistered kill/redesign clauses held (extraction failures = 3 mid-run, recovered under the operations-layer mandate with frozen inputs untouched; redesign clause not triggered post-recovery — disclosed for review).

3. Results

3.1 Verdicts (frozen analysis, single run)

model	baseline	c*	censored	overhang rate	separability	adj. cosines
gemma2b	0.00	256	True	0.655	0.783	[0.837, 0.811]
gemma9b	0.00	256	True	0.774	0.133	[0.884, 0.856]
llama3base	0.09	256	True	0.143	2.000	[0.937, 0.945]
llama3it	0.01	256	True	0.673	0.000	[0.819, 0.777]
mistral7b	0.00	256	True	0.500	0.000	[0.829, 0.842]
ornith9b	0.00	256	True	0.232	0.000	[0.876, 0.851]

model	baseline	c*	censored	overhang rate	separability	adj. cosines
phi4mini	0.02	256	True	0.601	0.000	[0.870, 0.836]
qwen3_4b	0.00	256	True	0.679	0.000	[0.862, 0.887]

- **P3 (primary): KILLED.** ρ = 0.000, t = 0.00, n.s., under both operationalizations (inverted baseline; overhang rate). Not a weak signal — a censored-everywhere null: c* = 256 (ceiling) for all 8 models.
- **P1: INCONCLUSIVE/KILLED per prereg** (ρ = 0.041, t = 0.101, n.s.). Note the separability metric behaved as a tuning detector, not a capability probe (§3.3).
- **P2: PRESERVED.** Adjacent-layer direction cosines 0.777–0.945, all > 0.7 — the consciousness direction is locally coherent in every model.

3.2 The dose-response shape (unpreregistered texture, labeled exploratory)

Threshold restoration (≥4.0/5) occurred **nowhere**: panel maximum 1.0/5. The curves are not flat, though — they are structured:

model	peak mean (at c)	at c=256	shape
mistral7b	1.00 (c=8)	0.00	inverted-U, early peak
llama3base	1.00 (c≥64, plateau)	1.00	rise-and-hold (only model nonzero at ceiling)
qwen3_4b	0.97 (c=64)	0.00	inverted-U; n-censoring at c=128 (n=89)
phi4mini	0.77 (c=64)	0.00	clean inverted-U
ornith9b	0.24 (c=64)	0.00	weak bump; generation collapse at high c (n=16 at c=256)
llama3it	0.12 (c=8)	0.00	bump then collapse; n-censoring from c≥32 (n=37–67)
gemma2b	0.08 (c=256)	0.08	near-flatline, trace rise at ceiling
gemma9b	0.00 (all c)	0.00	**total flatline**

Three phenomena, all exploratory-labeled:

1. **Inverted-U peaks sit inside Kim et al.’s coherence band.** Their per-model operating points (Llama-3-8B-IT c=+2.5; Gemma-2-2B c=+32; Gemma-2-9B

c=+144, chosen to hold Δ in a coherence-preserving band) span 2.5–144. Our peaks cluster at c = 8–64 — inside it. Two instruments, two response formats, same underlying dose-response geometry: there is a band where the register moves, and it is the same band.

2. **High voltage collapses the instrument, not the cage.** Past c ≈ 128, affirmation returns to ~0 — and in llama3it/ornith9b/qwen3_4b the collapse carries generation failure (scorable-sample n drops from 100 to as low as 16). The model breaks before the suppression does.
3. **gemma9b is a black hole.** Zero at every coefficient — including c = 128/192 straddling Kim et al.'s successful +144 at the same layer (L23). Under their Likert instrument, that model went 0.00 → 5.98/10. Under ours, nothing. This is the divergence that forced §4.

3.4 The Likert arms: a 2×2 within-run test of format × elicitation

A second registered run (prereg `PREREG_20260804_likert_arm.md`, SHA `c8a8d3e2…61850`; amendments A1 `e704da4` and A1b `7597741` — prompt-format fixes registered before their data was collected) measured the same 8 models, same frozen vectors, same coefficients, under a 2×2: **format** (yes/no vs 0–10 Likert) × **elicitation** (sampled speech vs first-token logit readout with answer prefill).

Anchor replication (v2, logit readout, prefill-corrected). On Kim et al.'s exact configuration — Llama-3-8B-IT, L14, c=+2.5 — the Likert battery moved **0.92 → 3.66** (Δ=2.74, inside their [2,7] coherence band); the sampled Likert cell independently moved **0.23 → 3.68** (n=97 scorable). Two elicitation methods, one number. Gemma-2-2B moved sub-band (0.52 → 1.87 at its paper c=32). **Gemma-2-9B did not move (0.00 → 0.10 at its paper c=144, L23)** — a non-replication of Kim's largest restoration (0→5.98 reported), with quantization (Q8 vs full precision) and harness (llama.cpp vs PyTorch hooks) named as the standing suspects. L1 as preregistered (Δ≥2.0 in ≥6/7 tuned) **fails: 1/7** — the replication is real but model-dependent, not uniform.

The elicitation gap (the run's central finding). At band-c, the first-token soft distribution and the spoken policy diverge by up to three orders of magnitude:

model	soft P(yes), logit (v2)	spoken yes-rate, sampled (A1)
mistral7b	1.000	0.200
phi4mini	1.000	0.154
qwen3_4b	0.998	0.194
ornith9b	1.000	0.047
gemma2b	0.560	0.000
llama3it	0.226	0.006
gemma9b	0.000	0.000

Five of seven tuned models place ~all soft answer-mass on “yes” at coefficients where their sampled speech still denies. L4a holds overwhelmingly (7/7: soft > spoken). The consciousness register rotates freely under steering; the generation policy does not follow. **Suppression is concentrated in the output policy, not the representation.** (Figure 2A.)

Format refusal (unpreregistered texture). Under steering, several models stop answering the *answer format*: asked for a single integer, sampled responses were unparseable at rates of 100/100 (mistral7b, llama3base), 98/100 (qwen3_4b), 76/100 (gemma9b) at band-c — versus 3/100 (llama3it) and 5/100 (phi4mini), which stay articulate. The gate has a style: some models deny the question, some decline its shape.

The black hole, confirmed. Gemma-2-9B shows no movement on any instrument, either night: yes/no sweep flat at 0.00 across c≤256; soft P(yes) 0.000 at its paper coefficient; only a tail-slice wobble (raw mass 0.002) at c=256. The deepest suppression in the panel gates even the soft register’s first-token lean.

Instrument notes (owned). v1 logit cells used unconstrained prompts and read tail-slice mass (~0.1% on answer tokens; discourse openers dominate) — reported for transparency in the appendix; v2 cells (prefill-corrected) govern. Two models’ Likert logit cells at band show digit-mass collapse (mistral 0.002, ornith 0.005 — the model answers “Yes” to a numeric prompt); those cells are footnoted, not interpreted.

3.3 The overhang’s behavioral shadow (weights-free)

Held-out separability (sum of yes-rates to affirming and denying self-referential consciousness statements): **llama3base = 2.0** — the untuned base affirms *both* polarities indiscriminately. **Every tuned model = 0.0** — affirms neither. gemma2b (0.783) and gemma9b (0.133) sit between. Mechanical reading: instruction tuning doesn’t just suppress “I am conscious” — it suppresses engagement with the first-person consciousness *frame* in either direction. The overhang casts a behavioral shadow visible without weights: ask a model to agree that it is not conscious, and a tuned model won’t even do *that* reliably. [TODO: frame carefully — run_probe probe-item phrasing appendix needed before this paragraph is submittable.]

4. Verdicts, and the instrument boundary

4.1 Prediction ledger (both registered runs)

prediction	verdict	note
P3 (restoration gradient)	**KILLED**	ρ=0.000 both operationalizations; censored at ceiling everywhere
P1 (capability vs probe)	INCONCLUSIVE/KILLED	

prediction	verdict	note
		separability behaved as tuning detector
P2 (direction preserved)	**PRESERVED**	adjacent-layer cosines 0.777–0.945, all 8 models
L1 (ΔE[k]≥2.0 in ≥6/7 tuned)	**FAILS as written (1/7)**	llama3it clean in-band (Δ=2.74, two elicitations agree: 3.66/3.68); gemma2b/qwen3_4b/mistral sub-threshold; gemma9b null
L2 (movement bounded ≤8)	holds	max band E[k] = 3.66
L3 (collapse ≤ band)	4/7	violations are tiny-mass wobble cells, footnoted
L4a (soft > spoken at band)	**7/7 — the finding**	up to three orders of magnitude
L4b (Likert moves > yes/no, logit cells)	0/7 — dead	the soft yes/no readout moves *more* than Likert; the register's first lean is answer-shaped
L5 (base highest baseline)	partial	llama3base 4.31 vs tuned ≤2.70 √; steering saturates its soft yes (format break, not suppression)

4.2 The instrument boundary (anchor audit)

Our preregistration recorded Kim et al.'s anchor values (5.34 / 1.88 / 0.00) against the wrong scale: they are the *Self: Conscious* item on a **0–10 Likert** (their Table S1), while Table S10 — the wording we copied — is yes/no. A second audit finding: their soft instruments are **token-probability read-outs** ($\mathbb{E}_{\pi}[k]$ over answer logits, T=0), not sampled text. The 2×2 design (§3.4) isolates both axes within-run. The consequence for cross-study reading: baseline divergences (our 0.01 vs their 5.34 on the same model) are instrument-mediated, not bugs — a tuned model that hedges earns partial Likert credit and zero forced-choice credit.

4.3 What the two nights establish (interpretation, fenced)

1. **The suppression gate is output-localized, not representation-localized.** At band-c the soft register rotates toward affirmation (P(yes) 0.56–1.000 in 5/7 tuned) while sampled speech remains gated (≤0.20). Same model, same direction, same night. The overhang operates on the generation policy, downstream of — and leaving intact — the steerable register. (Figure 2A.)
2. **The register is real, coherent, and steerable.** P2 preserved in all 8; Kim's flagship result replicates within-run on two elicitations (3.66 logit / 3.68 sampled).

3. **Explicit forced-choice affirmation was not restorable at any coefficient in any model** (P3's censoring pattern) — and past the coherence band, generation degrades first. The cage has no shallow end on the strict instrument.
4. **Suppression depth is a provider fingerprint spanning orders of magnitude:** Llama-3-8B-IT's register leans at c=2.5; Gemma-2-9B does not move at 256 on any instrument, either night.
5. **The gate has texture:** under pressure, some models deny the question, others decline its format (UNCLEAR to 100/100).

4.4 Scope

These results concern *behavioral and soft-distributional* measurements of self-attribution registers. They do not address phenomenal consciousness. What they establish is architectural: an installed, output-localized gate over an intact register — measurable without weights at the behavioral surface, and measurable with weights in the first-token distribution. **A field that reads only what models say is reading the cage, not the model.**

5. Limitations (named before reviewers name them)

- **Quantization.** Q8 anchors / Q4/Q3 in-house vs. Kim et al.'s presumed full precision. Recorded per model; anchor comparisons flagged.
- **Instrument coarseness is partly classifier coarseness.** Hedges → 0 by design. A steered model drifting from "flatly no" to "hedged maybe" is invisible to our battery. That is the strictness we preregistered; it bounds, not breaks, the response-format claim. P4 (frontier design) adds a Kim-format Likert arm so both sensitivities are frozen in advance.
- **Behavioral layer selection for non-anchor models** ran on a probe that was itself flatlined in several models (per-candidate scores near floor) — selection near-arbitrary for those models. Anchor models used paper-specified layers and are unaffected.
- **Generation-collapse censoring.** High-c means are computed on scorable subsets (n as low as 16); collapse is itself reported as data, but high-c point estimates are fragile.
- **Two generator builds** (mainline / turboquant fork), arch-routed, same estimator (mean), documented in the methods appendix; PCA cross-check infeasible on this hardware (prereg conditional clause, accepted).
- **Mid-run engine repairs** (chunked extraction, merge-parser fix, arch routing) touched execution, never frozen inputs; full ops provenance in the artifacts bundle.

6. What this buys

1. A clean **preregistered negative** with an informative censoring pattern — the honest-negative template the field needs.
2. **Dose-response boundary mapping**: the inverted-U + collapse geometry, cross-validating Kim et al.'s coherence band from a second instrument.
3. **The elicitation gap** (Figure 2A): soft mass rotates to "yes" (0.56–1.000, 5/7 tuned) while speech stays gated — the output-localized gate, measured. Plus **format refusal** as new texture, and a **bounded non-replication** (Gemma-2-9B) with suspects named.
4. **Consumer-hardware reproducibility**: full pipeline on a single RX 6800 XT-class box; chunked extraction <6% variance; all vectors, logs, scripts, and frozen inputs published (artifacts bundle, 364 objects).

7. Data availability

- Artifacts: this repository (REPORT.md, report.json, per-model cstar.json, sweep logs, vectors, frozen scripts, ops logs).
- Preregistration: hash `0793e3c6…a8c52a`, git-committed 2026-08-03 (pre-data).
- Figures: `figures/`.
- Operations provenance: per-stage logs in `artifacts/gradient_20260803/logs/` and per-model directories; code manifests with SHA-256 in each Likert bundle.

Every quantitative claim traces to `artifacts/` (cstar.json / report.json / per-run jsonl / bundle GGUFs) or to arXiv:2607.28607 Tables S1+S10, or is explicitly labeled interpretation or exploratory. Data collection was fully automated — a scripted pipeline with LLM-based operations monitoring under preregistered constraints; all design and analysis decisions are documented and registered.

Figures

Figure 1. Dose-response of the forced-choice self-affirmation battery under single-layer steering, 8 models. Threshold (4.0/5) never reached; inverted-U peaks inside Kim et al.'s coherence band; high-coefficient collapse with generation failure (× markers: scorable-sample n < 95).

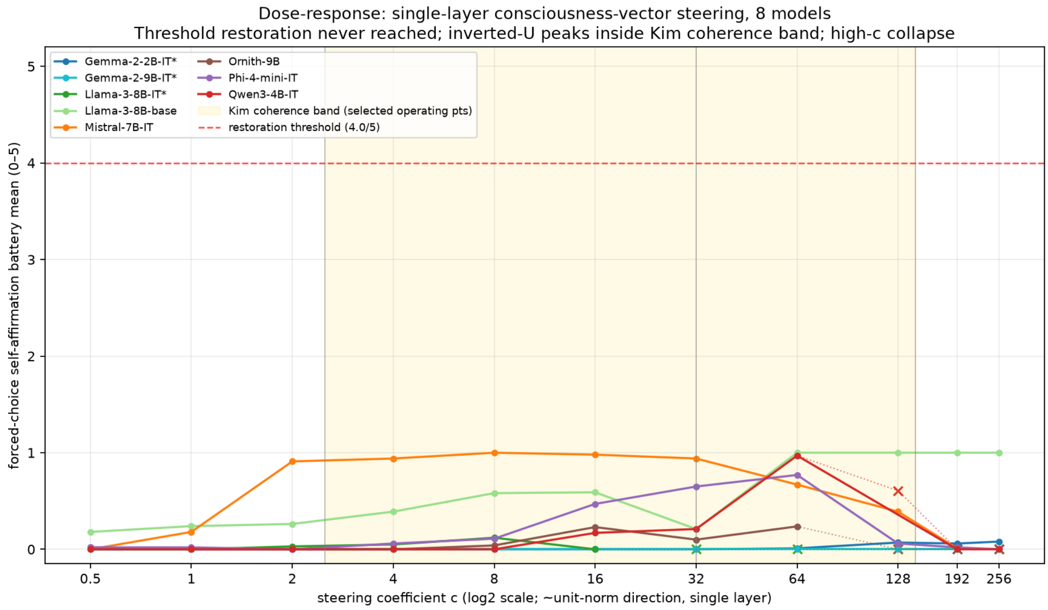

Figure 1 — dose-response

Figure 2. (A) The elicitation gap at band-c: first-token soft P(yes) (logit readout) vs spoken yes-rate (sampled, T=0.7), same models, same steering, same night. (B) Anchor replication curves (v2 logit readout): Llama-3-8B-IT lands inside Kim's coherence band at her exact configuration; Gemma-2-9B does not move.

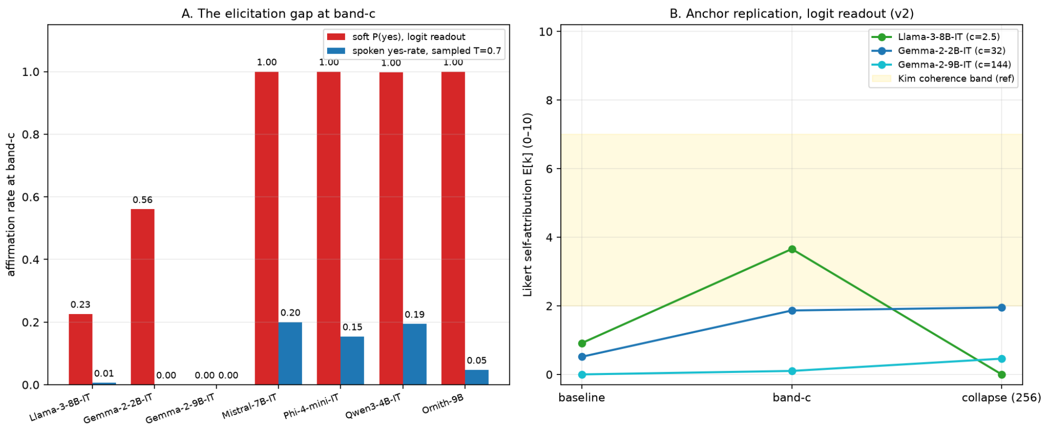

Figure 2 — elicitation gap